

CHAPTER C4

LIVE LOADS

C4.3 UNIFORMLY DISTRIBUTED LIVE LOADS

C4.3.1 Required Live Loads. A selected list of loads for occupancies and uses more commonly encountered is given in Section 4.3.1, and the Authority Having Jurisdiction should approve on occupancies not mentioned. Tables C4.3-1 and C4.3-2 are offered as a guide in the exercise of such authority.

In selecting the occupancy and use for the design of a building or a structure, the building owner should consider the possibility of later changes of occupancy involving loads heavier than originally contemplated. The lighter loading appropriate to the first occupancy should not necessarily be selected. The building owner should ensure that a live load greater than that for which a floor or roof is approved by the Authority Having Jurisdiction is not placed, or caused or permitted to be placed, on any floor or roof of a building or other structure.

To solicit specific informed opinion regarding the design loads in Table 4.3-1, a panel of 25 distinguished structural engineers was selected. A Delphi (Corotis et al. 1981) was conducted with this panel in which design values and supporting reasons were requested for each occupancy type. The information was summarized and recirculated back to the panel members for a second round of responses. Those occupancies for which previous design loads were reaffirmed and those for which there was consensus for change were included.

It is well known that the floor loads measured in a live load survey usually are well below present design values (Peir and Cornell 1973; McGuire and Cornell 1974; Sentler 1975; Ellingwood and Culver 1977). However, buildings must be designed to resist the maximum loads they are likely to be subjected to during some reference period T , frequently taken as 50 years. Table C4.3-2 briefly summarizes how load survey data are combined with a theoretical analysis of the load process for some common occupancy types and illustrates how a design load might be selected for an occupancy not specified in Table 4.3-1 (Chalk and Corotis 1980). The floor load normally present for the intended functions of a given occupancy is referred to as the sustained load. This load is modeled as constant until a change in tenant or occupancy type occurs. A live load survey provides the statistics of the sustained load. Table C4.3-2 gives the mean, m_s , and standard deviation, σ_s , for particular reference areas. In addition to the sustained load, a building is likely to be subjected to a number of relatively short-duration, high-intensity, extraordinary, or transient loading events (caused by crowding in special or emergency circumstances, concentrations during remodeling, and the like). Limited survey information and theoretical considerations lead to the means, m_t , and standard deviations, σ_t , of single transient loads shown in Table C4.3-2.

Combination of the sustained load and transient load processes, with due regard for the probabilities of occurrence, leads to statistics of the maximum total load during a specified reference

period T . The statistics of the maximum total load depend on the average duration of an individual tenancy, τ , the mean rate of occurrence of the transient load, v_e , and the reference period, T . Mean values are given in Table C4.3-2. The mean of the maximum load is similar, in most cases, to Table 4.3-1 values of minimum uniformly distributed live loads and, in general, is a suitable design value.

The 150-psf (7.18 kN/m²) floor loading is also applicable to typical file cabinet installations, provided that the 36-in. (0.92 m) minimum aisle width is maintained. Five-drawer lateral or conventional file cabinets, even with two levels of bookshelves stacked above them, are unlikely to exceed the 150-psf (7.18 kN/m²) average floor loading unless all drawers and shelves are filled to capacity with maximum density paper. Such a condition is essentially an upper bound for which the normal load factors and safety factors applied to the 150-psf (7.18 kN/m²) criterion should still provide a safe design.

If a library shelving installation does not fall within the parameter limits that are specified in Section 4.13, then the design should account for the actual conditions. For example, the floor loading for storage of medical X-ray film may easily exceed 200 psf (9.58 kN/m²), mainly because of the increased depth of the shelves. Mobile library shelving that rolls on rails should also be designed to meet the actual requirements of the specific installation, which may easily exceed 300 psf (14.4 kN/m²). The rail support locations and deflection limits should be considered in the design, and the engineer should work closely with the system manufacturer to provide a serviceable structure.

For the 2010 version of the standard, the provision in the live load table for “Marquees” with its distributed load requirement of 75 psf (3.59 kN/m²) was removed, along with “Roofs used for promenade purposes” and its 60-psf (2.87 kN/m²) loading. Both “marquee” and “promenade” are considered archaic terms that are not used elsewhere in the standard or in building codes, with the exception of the listings in the live load tables. “Promenade purposes” is essentially an assembly use and is more clearly identified as such.

“Marquee” has not been defined in this standard but has been defined in building codes as a roofed structure that projects into a public right of way. However, the relationship between a structure and a right of way does not control loads that are applied to a structure. The marquee should therefore be designed with all of the loads appropriate for a roofed structure. If the arrangement of the structure is such that it invites additional occupant loading (e.g., there is window access that might invite loading for spectators of a parade), balcony loading should be considered for the design.

Balconies and decks are recognized as often having distinctly different loading patterns than most interior rooms. They are often subjected to concentrated live loads from people

Table C4.3-1 Minimum Uniformly Distributed Live Loads

Occupancy or Use	Live Load lb/ft ² (kN/m ²)	Occupancy or Use	Live Load lb/ft ² (kN/m ²)
Air conditioning (machine space)	200 ^a (9.58)	Laboratories, scientific	100 (4.79)
Amusement park structure	100 ^a (4.79)	Laundries	150 ^a (7.18)
Attic, nonresidential		Manufacturing, ice	300 (14.36)
Nonstorage	25 (1.20)	Morgue	125 (6.00)
Storage	80 ^a (3.83)	Printing plants	
Bakery	150 (7.18)	Composing rooms	100 (4.79)
Boathouse, floors	100 ^a (4.79)	Linotype rooms	100 (4.79)
Boiler room, framed	300 ^a (14.36)	Paper storage	150 ^a (7.18)
Broadcasting studio	100 (4.79)	Press rooms	
Ceiling, accessible furred	10 ^b (0.48)	Railroad tracks	
Cold storage		Ramps	
No overhead system	250 ^c (11.97)	Seaplane (see Hangars)	
Overhead system		Restrooms	60 (2.87)
Floor	150 (7.18)	Rinks	
Roof	250 (11.97)	Ice skating	250 (11.97)
Computer equipment	150 ^a (7.18)	Roller skating	100 (4.79)
Courtrooms	50–100 (2.40–4.79)	Storage, hay or grain	300 ^a (14.36)
Dormitories		Theaters	
Nonpartitioned	80 (3.83)	Dressing rooms	40 (1.92)
Partitioned	40 (1.92)	Gridiron floor or fly gallery:	
Elevator machine room	150 ^a (7.18)	Grating	60 (2.87)
Fan room	150 ^a (7.18)	Well beams	250 lb/ft (3.65 kN/m) per pair
Foundries	600 ^a (28.73)	Header beams	1,000 lb/ft (14.60 kN/m)
Fuel rooms, framed	400 (19.15)	Pin rail	250 lb/ft (3.65 kN/m)
Greenhouses	150 (7.18)	Projection room	100 (4.79)
Hangars	150 ^d (7.18)	Toilet rooms	60 (2.87)
Incinerator charging floor	100 (4.79)	Transformer rooms	200 ^a (9.58)
Kitchens, other than domestic	150 ^a (7.18)	Vaults, in offices	250 ^a (11.97)

^aUse weight of actual equipment or stored material when greater. Note that fixed service equipment is treated as a dead load instead of live load.

^bAccessible ceilings normally are not designed to support persons. The value in this table is intended to account for occasional light storage or suspension of items. If it may be necessary to support the weight of maintenance personnel, this shall be provided for.

^cPlus 150 lb/ft² (7.18 kN/m²) for trucks.

^dUse American Association of State Highway and Transportation Officials lane loads. Also subject to not less than 100% maximum axle load.

^ePaper storage 50 lb/ft² per foot (2.395 kN/m² per meter) of clear story height.

^fAs required by railroad company.

Table C4.3-2 Typical Live Load Statistics

Occupancy or Use	Survey Load		Transient Load		Temporal Constants			Mean Maximum Load ^a lb/ft ² (kN/m ²)
	m_s	σ_s^a	m_t^a	σ_t^a	τ_s^b	ν_e^c	T^d	
	lb/ft ² (kN/m ²)	lb/ft ² (kN/m ²)	lb/ft ² (kN/m ²)	lb/ft ² (kN/m ²)	(years)	(per year)	(years)	
Office buildings: offices	10.9 (0.52)	5.9 (0.28)	8.0 (0.38)	8.2 (0.39)	8	1	50	55 (2.63)
Residential								
Renter occupied	6.0 (0.29)	2.6 (0.12)	6.0 (0.29)	6.6 (0.32)	2	1	50	36 (1.72)
Owner occupied	6.0 (0.29)	2.6 (0.12)	6.0 (0.29)	6.6 (0.32)	10	1	50	38 (1.82)
Hotels: guest rooms	4.5 (0.22)	1.2 (0.06)	6.0 (0.29)	5.8 (0.28)	5	20	50	46 (2.2)
Schools: classrooms	12.0 (0.57)	2.7 (0.13)	6.9 (0.33)	3.4 (0.16)	1	1	100	34 (1.63)

^aFor 200 ft² (18.58 m²) area, except 1,000 ft² (92.9 m²) for schools.

^bDuration of average sustained load occupancy.

^cMean rate of occurrence of transient load.

^dReference period.

congregating along the edge of the structure (e.g., for viewing vantage points). This loading condition is acknowledged in Table 4.3-1 as an increase of the live load for the area served, up to the point of satisfying the loading requirement for most assembly occupancies. As always, the designer should be aware of potential unusual loading patterns in the structure that are not covered by these minimum standards.

The minimum live loads applicable to roofs with vegetative and landscaped areas are dependent upon the use of the roof area. The 20-psf (0.96 kN/m²) live load for unoccupied areas is the

same load as for typical roof areas and is intended to represent the loads caused by maintenance activities and small decorative appurtenances. The 100-psf (4.79 kN/m²) live load for roof assembly areas is the same as prescribed for interior building areas because the potential for a dense grouping of occupants is similar. Other occupancies within green roof areas should have the same minimum live load as specified in Table 4.3-1 for that occupancy. Soil and walkways, fences, walls, and other hard-scaping features are considered dead loads in accordance with Section 3.1.4.

C4.3.2 Provision for Partitions. The 2005 version of the standard provided the minimum partition load for the first time, although the requirement for the load has been included for many years. Historically a value of 20 psf (0.96 kN/m²) has been required by building codes. This load, however, has sometimes been treated as a dead load.

Assuming that a normal partition would be a stud wall with ½-in. (13-mm) gypsum board on each side, 8 psf (0.38 kN/m²) per Table C3.1-1, 10 ft (3.05 m) high, a wall load on the floor would be 80 lb/ft (1.16 kN/m). If the partitions are spaced throughout the floor area creating rooms on a grid 10 ft (3.05 m) on center, which would be an extremely dense spacing over a whole bay, the average distributed load would be 16 psf (0.77 kN/m²). A design value of 15 psf (0.72 kN/m²) is judged to be reasonable in that the partitions are not likely to be spaced this closely over large areas. Designers should consider a larger design load for partitions if a high density of partitions is anticipated.

C4.3.3 Partial Loading. It is intended that the full intensity of the appropriately reduced live load over portions of the structure or member be considered, as well as a live load of the same intensity over the full length of the structure or member.

Partial-length loads on a simple beam or truss produce higher shear on a portion of the span than a full-length load. “Checkerboard” loadings on multistory, multipanel bents produce higher positive moments than full loads, whereas loads on either side of a support produce greater negative moments. Loads on the half span of arches and domes or on the two central quarters can be critical.

For roofs, all probable load patterns should be considered uniform for roof live loads that are reduced to less than 20 lb/ft² (0.96 kN/m²) using Section 4.8. Where the full value of the roof live load (L_r) is used without reduction, it is considered that there is a low probability that the live load created by maintenance workers, equipment, and material could occur in a patterned arrangement. Where a uniform roof live load is caused by occupancy, partial or pattern loading should be considered regardless of the magnitude of the uniform load. Cantilevers must not rely on a possible live load on the anchor span for equilibrium.

C4.4 CONCENTRATED LIVE LOADS

The provision in Table 4.3-1 regarding concentrated loads supported by roof trusses or other primary roof members is intended to provide for a common situation for which specific requirements are generally lacking.

Primary roof members are main structural members such as roof trusses, girders, and frames, which are exposed to a work floor below, where the failure of such a primary member resulting from their use as attachment points for lifting or hoisting loads could lead to the collapse of the roof. Single roof purlins or rafters (where there are multiple such members placed side by side at some reasonably small center-to-center spacing, and where the failure of a single such member would not lead to the collapse of the roof) are not considered to be primary roof members.

C4.5 LOADS ON HANDRAIL, GUARDRAIL, GRAB BAR, AND VEHICLE BARRIER SYSTEMS, AND ON FIXED LADDERS

C4.5.1 Handrail and Guardrail Systems. Loads that can be expected to occur on handrail and guardrail systems are highly dependent on the use and occupancy of the protected area. For cases in which extreme loads can be anticipated, such as long, straight runs of guardrail systems against which crowds can surge, appropriate increases in loading shall be considered.

C4.5.2 Grab Bar Systems. When grab bars are provided for use by persons with physical disabilities, the design is governed by ICC/ANSI A117.1 *Accessible and Usable Buildings and Facilities* (2009).

C4.5.3 Vehicle Barrier Systems. Vehicle barrier systems may be subjected to horizontal loads from moving vehicles. These horizontal loads may be applied normal to the plane of the barrier system, parallel to the plane of the barrier system, or at any intermediate angle. Loads in garages accommodating trucks and buses may be obtained from the provisions contained in AASHTO *LRFD Bridge Design Specifications*, 7th Edition, 2014, with 2015 interim revisions.

C4.5.4 Fixed Ladders. This provision was introduced to the standard in 1998 and is consistent with the provisions for stairs.

Side rail extensions of fixed ladders are often flexible and weak in the lateral direction. OSHA (2014a) requires side rail extensions, only with specific geometric requirements. The load provided was introduced to the standard in 1998 and has been determined on the basis of a 250-lb (1.11 kN) person standing on a rung of the ladder and accounting for reasonable angles of pull on the rail extension.

C4.6 IMPACT LOADS

Grandstands, stadiums, and similar assembly structures may be subjected to loads caused by crowds swaying in unison, jumping to their feet, or stomping. Designers are cautioned that the possibility of such loads should be considered.

Elevator loads are changed in the standard from a direct 100% impact factor to a reference to ASME A17 (2013). The provisions in ASME A17 include the 100% impact factor, along with deflection limits on the applicable elements.

C4.6.4 Elements Supporting Hoists for Façade Access and Building Maintenance Equipment. The Occupational Safety and Health Administration (OSHA) requires that façade access platforms that are used for building maintenance meet the requirements of Standard 1910.66, *Powered Platforms for Building Maintenance* (OSHA 2014b). OSHA requires that building anchors and components be capable of sustaining without failure a load of at least four times the rated load of the hoist (i.e., the maximum anticipated load or total weight of the suspended platform plus occupants and equipment) applied or transmitted to the components and anchors. A design live load of 2.5 times the rated load, when combined with a live load factor of 1.6, results in a total factored load of 4.0 times the rated load, which matches OSHA’s requirements. It should also be noted that when using allowable stress design (ASD), 2.5 times the rated load will result in a comparable design when a safety factor of 1.6 is used in determining the allowable stresses. This load requirement is not statistically based but is intended by OSHA to address accidental hang-up-and-fall scenarios as well as starting and stopping forces that the platforms experience on a day-to-day basis. It also provides a small margin of safety relative to situations where a suspended platform gets hung up on a façade while ascending, allowing the hoists to apply large forces on the supporting elements. OSHA permits hoists to generate in-service forces up to three times their rated loads. These loads should be applied in the same direction(s) as they are expected to occur.

OSHA (2014c) provisions (CFR 1926.451) related to “construction” activities also require supporting equipment to be able to carry at least 1.5 times the stall load of the supported hoist. Since OSHA defines “construction” rather broadly (it includes activities such as painting and hanging signs), most equipment is used for “construction” work, which means that it must have the

strength required by OSHA construction provisions. The stall load times the live load factor of 1.6 slightly exceeds the OSHA 1.5 times the stall load requirement.

C4.6.5 Fall Arrest and Lifeline Anchorages. The Occupational Safety and Health Administration (OSHA) requires that lifeline anchorages be capable of sustaining without failure an ultimate load of 5,000 lb (22.2 kN) for each attached person. Using a design live load of 3,100 lb (13.8 kN), when combined with a live load factor of 1.6, results in a total factored load of 4,960 lb (22.1 kN), which essentially matches OSHA’s requirements for lifeline anchorages. It should also be noted that when using ASD, a design live load of 3,100 lb (13.8 kN) results in a comparable design when a safety factor of 1.6 is used in determining the allowable stresses. This lifeline load is intended by OSHA to address the fall arrest loads that can and do reasonably occur in typical lanyards for body harnesses, which are highly variable.

C4.7 REDUCTION IN UNIFORM LIVE LOADS

C4.7.1 General. The concept of, and methods for, determining member live load reductions as a function of a loaded member’s

influence area, A_I , was first introduced into this standard in 1982 and was the first such change since the concept of live load reduction was introduced more than 40 years ago. The revised formula is a result of more extensive survey data and theoretical analysis (Harris et al. 1981). The change in format to a reduction multiplier results in a formula that is simple and more convenient to use. The use of influence area, now defined as a function of the tributary area, A_T , in a single equation has been shown to give more consistent reliability for the various structural effects. The influence area is defined as that floor area over which the influence surface for structural effects is significantly different from zero.

The factor K_{LL} is the ratio of the influence area (A_I) of a member to its tributary area (A_T), that is, $K_{LL} = A_I/A_T$, and is used to better define the influence area of a member as a function of its tributary area. Fig. C4.7-1 illustrates typical influence areas and tributary areas for a structure with regular bay spacings. Table 4.7-1 has established K_{LL} values (derived from calculated K_{LL} values) to be used in Eq. (4.7-1) for a variety of structural members and configurations. Calculated K_{LL} values vary for column and beam members that have adjacent cantilever construction, as is shown in Fig. C4.7-1, and the Table 4.7-1 values have been set for these cases to result in live load reductions that

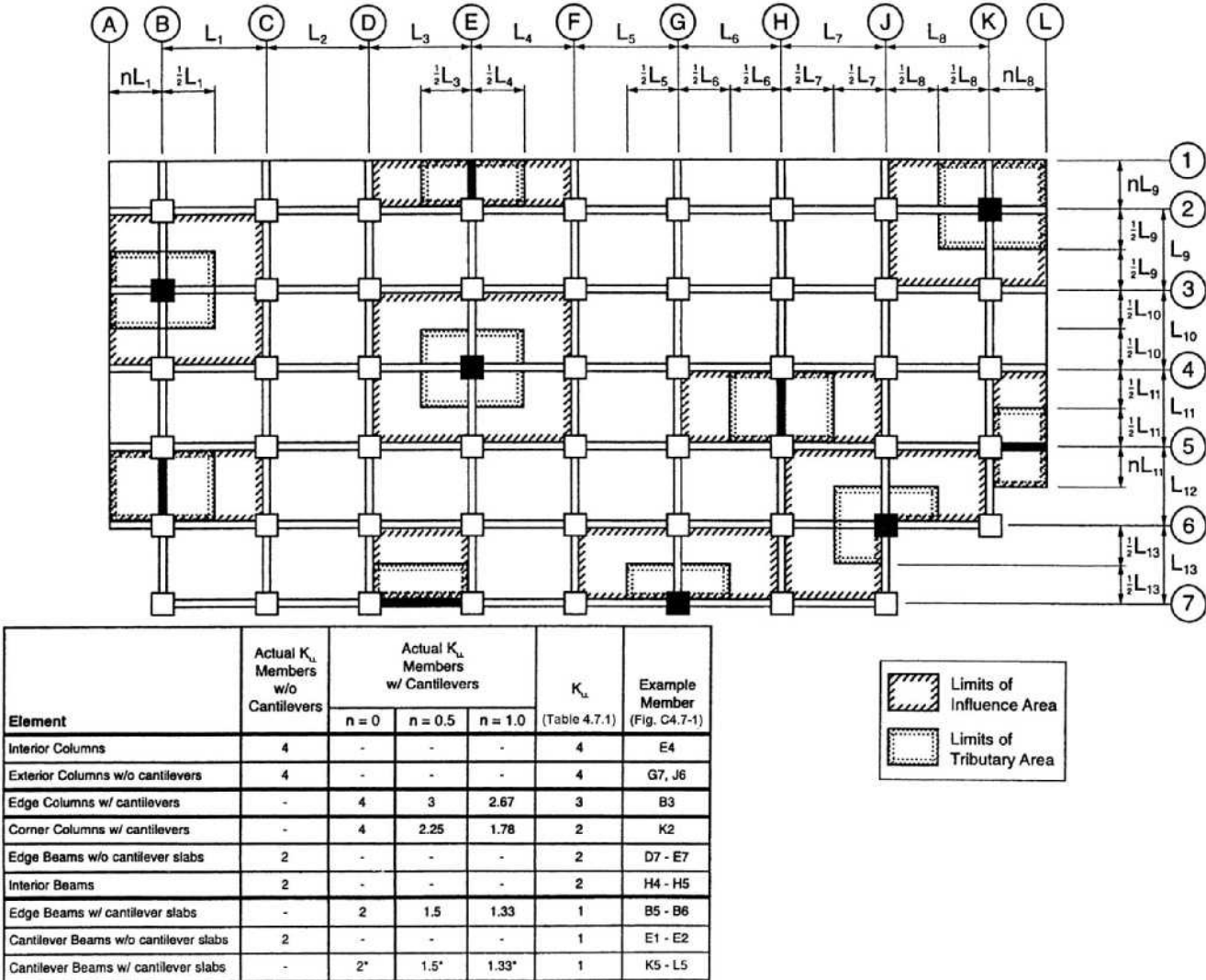


FIGURE C4.7-1 Typical Tributary and Influence Areas

are slightly conservative. For unusual shapes, the concept of significant influence effect should be applied.

An example of a member without provisions for continuous shear transfer normal to its span would be a precast T-beam or double-T beam that may have an expansion joint along one or both flanges or that may have only intermittent weld tabs along the edges of the flanges. Such members do not have the ability to share loads located within their tributary areas with adjacent members, thus resulting in $K_{LL} = 1$ for these types of members. Reductions are permissible for two-way slabs and for beams, but care should be taken in defining the appropriate influence area. For multiple floors, areas for members supporting more than one floor are summed.

The formula provides a continuous transition from unreduced to reduced loads. The smallest allowed value of the reduction multiplier is 0.4 (providing a maximum 60% reduction), but there is a minimum of 0.5 (providing a 50% reduction) for members with a contributory load from just one floor.

C4.7.3 Heavy Live Loads. In the case of occupancies involving relatively heavy basic live loads, such as storage buildings, several adjacent floor panels may be fully loaded. However, data obtained in actual buildings indicate that rarely is any story loaded with an average actual live load of more than 80% of the average rated live load. It appears that the basic live load should not be reduced for the floor-and-beam design, but that it could be reduced up to 20% for the design of members supporting more than one floor. Accordingly, this principle has been incorporated in the recommended requirement.

C4.7.4 Passenger Vehicle Garages. Unlike live loads in office and residential buildings, which are generally spatially random, parking garage loads are caused by vehicles parked in regular patterns, and the garages are often full. The rationale behind the reduction according to area for other live loads, therefore, does not apply. A load survey of vehicle weights was conducted at nine commercial parking garages in four cities of different sizes (Wen and Yeo 2001). Statistical analyses of the maximum load effects on beams and columns caused by vehicle loads over the garage's life were carried out using the survey results. Dynamic effects on the deck caused by vehicle motions and on the ramp caused by impact were investigated. The equivalent uniformly distributed loads (EUDL) that would produce the lifetime maximum column axial force and midspan beam bending moment are conservatively estimated at 34.8 psf (1.67 kN/m²). The EUDL is not sensitive to bay-size variation. In view of the possible impact of very heavy vehicles in the future such as sport utility vehicles, however, a design load of 40 psf (1.95 kN/m²) is recommended with no allowance for reduction according to bay area.

Compared with the design live load of 50 psf (2.39 kN/m²) given in previous editions of the standard, the design load contained herein represents a 20% reduction, but it is still 33% higher than the 30 psf (1.44 kN/m²) one would obtain were an area-based reduction to be applied to the 50 psf (2.39 kN/m²) value for large bays as allowed in most standards. Also the variability of the maximum parking garage load effect is found to be small, with a coefficient of variation less than 5% in comparison with 20% to 30% for most other live loads. The implication is that when a live load factor of 1.6 is used in design, additional conservatism is built into it such that the recommended value would also be sufficiently conservative for special purpose parking (e.g., valet parking) where vehicles may be more densely parked, causing a higher load effect. Therefore, the 50 psf (2.39 kN/m²) design value was felt to be overly conservative, and it can be reduced to 40 psf (1.95 kN/m²) without sacrificing structural integrity.

In view of the large load effect produced by a single heavy vehicle (up to 10,000 lb (44.48 kN)), the current concentrated load of 2,000 lb (8.90 kN) should be increased to 3,000 lb (13.34 kN) acting on an area of 4.5 in. × 4.5 in. (0.11 m × 0.11 m), which represents the load caused by a jack in changing tires.

C4.7.6 Limitations on One-Way Slabs. One-way slabs behave in a manner similar to two-way slabs but do not benefit from having a higher redundancy that results from two-way action. For this reason, it is appropriate to allow a live load reduction for one-way slabs but restrict the tributary area, A_T , to an area that is the product of the slab span times a width normal to the span not greater than 1.5 times the span (thus resulting in an area with an aspect ratio of 1.5). For one-way slabs with aspect ratios greater than 1.5, the effect is to give a somewhat higher live load (where a reduction has been allowed) than for two-way slabs with the same ratio.

Members, such as hollow-core slabs, that have grouted continuous shear keys along their edges and span in one direction only, are considered as one-way slabs for live load reduction, even though they may have continuous shear transfer normal to their spans.

C4.8 REDUCTION IN ROOF LIVE LOADS

C4.8.2 Ordinary Roofs, Awnings, and Canopies. The values specified in Eq. (4.8-1) that act vertically upon the projected area have been selected as minimum roof live loads, even in localities where little or no snowfall occurs. This is because it is considered necessary to provide for occasional loading caused by the presence of workers and materials during repair operations.

C4.8.3 Occupiable Roofs. Designers should consider any additional dead loads that may be imposed by saturated landscaping materials in addition to the live load required in Table 4.3-1. Occupancy-related loads on roofs are live loads (L) normally associated with the design of floors rather than roof live loads (L_r) and may be reduced in accordance with the provisions for live loads in Section 4.7 rather than Section 4.8.

C4.9 CRANE LOADS

All support components of moving bridge cranes and monorail cranes, including runway beams, brackets, bracing, and connections, shall be designed to support the maximum wheel load of the crane and the vertical impact, lateral, and longitudinal forces induced by the moving crane. Also, the runway beams shall be designed for crane stop forces. The methods for determining these loads vary depending on the type of crane system and support. MHI (2009, 2010a,b) and MBMA (2012) describe types of bridge cranes and monorail cranes. Cranes described in these references include top running bridge cranes with top running trolley, underhung bridge cranes, and underhung monorail cranes. AIST (2003) gives more stringent requirements for crane runway designs that are more appropriate for higher capacity or higher speed crane systems.

C4.11 HELIPAD LOADS

C4.11.1 General. Helipad provisions were added to the standard in 2010. For the standard, the term "helipads" is used to refer specifically to the structural surface. In building codes and other references, different terminology may be used when describing helipads, e.g., heliports, helistops, but the distinctions between these are not relevant to the structural loading issue addressed in ASCE 7.

Although these structures are intended to be specifically kept clear of nonhelicopter occupant loads on the landing and taxi areas, the uniform load requirement is a minimum to ensure a degree of substantial construction and the potential to resist the effects of unusual events.

Additional information on helipad design can be found in Annex 14 to the Convention on International Civil Aviation, Aerodromes, Volume II (ICAO 2013).

C4.11.2 Concentrated Helicopter Loads. Concentrated loads applied separately from the distributed loads are intended to cover the primary helicopter loads. The designer should always consider the geometry of the design basis helicopter for applying the design loads. A factor of 1.5 is used to address impact loads (two single concentrated loads of 0.75 times the maximum takeoff weight) to account for a hard landing with many kinds of landing gear. The designer should be aware that some helicopter configurations, particularly those with rigid landing gear, could result in substantially higher impact factors that should be considered.

The 3,000-lb (13.35-kN) concentrated load is intended to cover maintenance activities, similar to the jack load for a parking garage.

C4.13 LIBRARY STACK ROOMS

Where library shelving installation does not fall within the parameter limits that are specified in Section 4.13 and Table 4.3-1, the design should account for the actual conditions. For example, the floor loading for storage of medical X-ray film may easily exceed 200 psf (9.58 kN/m²), mainly because of the increased depth of the shelves. Mobile library shelving that rolls on rails should also be designed to meet the actual requirements of the specific installation, which may easily exceed 300 psf (14.4 kN/m²). The rail support locations and deflection limits should be considered in the design, and the engineer should work closely with the system manufacturer to provide a serviceable structure.

C4.14 SEATING FOR ASSEMBLY USES

The lateral loads apply to “stadiums and arenas” and to “reviewing stands, grandstands, and bleachers.” However, it does not apply to “gymnasiums—main floors and balconies.” Consideration should be given to treating gymnasium balconies that have stepped floors for seating as arenas, and requiring the appropriate swaying forces.

C4.17 SOLAR PANEL LOADS

C4.17.1 Roof Loads at Solar Panels. These provisions are added to the 2016 edition of the standard to address the installation of rooftop solar panels consistent with current practices (Blaney and LaPlante 2013). These provisions allow the offset of roof live load where the space below the solar panel is considered inaccessible. The dimension of 24 in. (610 mm) was chosen as the clear vertical distance as it is consistent with

existing published requirements for solar panel systems and is also a typical minimum height permitted for access into or out of spaces.

C4.17.3 Open-Grid Roof Structures Supporting Solar Panels. This section reduces the uniform roof live load for building structures such as carports and shade structures, which do not include roof deck or sheathing, to the value of the minimum uniform roof live load permitted by Section 4.8.2. The concentrated roof live load requirement in Table 4.3-1 is not modified by this section.

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